

NLP Projects

General Description of the projects:

It is expected that the students use one of the following techniques; e.g.; RNN, LSTM, GRU, HMM, and/or transformers to solve one of the following NLP problems. It is also expected that you have a suitable dataset and split it into a training and testing dataset. Performance evaluation via computing the accuracy, confusion matrix, ...etc are required.

1. Sentiment analysis

This is one of the most popular NLP. The reason for its popularity is that it is widely used by companies to monitor the review of their product through customer feedback. If the review is mostly positive, the companies get an idea that they are on the right track. And, if the sentiment of the reviews concluded using this NLP Project are mostly negative then, the company can take steps to improve their product.

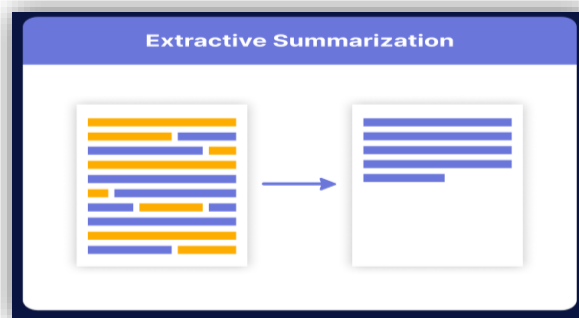
2. Chatbot

Most tech companies are now utilizing conversational bots, called Chatbots to interact with their customers and resolve their issues. This is a very good way of saving time for both customers and companies. The users are guided to first enter all the details that the bots ask for and only if there is a need for human intervention, the customers are connected with a customer care executive.

3. Text Summarization (Extractive Summarization)

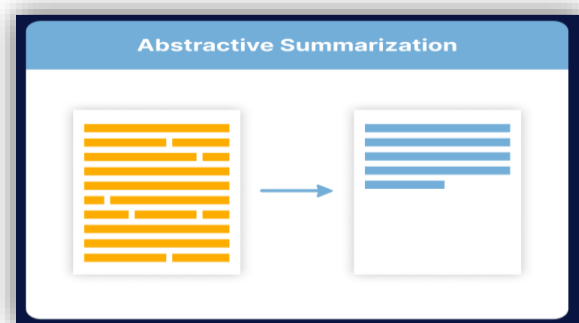
People now want everything to be given to them at a fast speed. And that is why short news articles are becoming more popular than long news articles. One such instance of this is the popularity of the Inshorts application that summarizes the lengthy news articles into just 60 words. And the app is able to achieve this by using NLP algorithms for text summarization. Extractive

summarization is the strategy of concatenating extracts taken from a corpus into a summary.



4. Text Summarization (Abstractive Summarization)

People now want everything to be given to them at a fast speed. And that is why short news articles are becoming more popular than long news articles. One such instance of this is the popularity of the Inshorts application that summarizes the lengthy news articles into just 60 words. Abstractive summarization involves paraphrasing the corpus using novel sentences.



5. Machine Translation

Many languages don't allow for straight translation and have different orders for sentence structure, which translation services used to overlook. With NLP, online translators can translate languages more accurately. This is infinitely helpful when trying to communicate with someone in another language. Not only that, but when translating from another language to your own, tools now recognize the language based on inputted text and translate it.

6. Email Filter

Recall those not-so-good old days of using emails where we used to receive so many junk emails and very few relevant emails. Using the NLP algorithms, email service providing systems can identify spam emails easily which helps their user base in saving time by avoiding unnecessary emails in their inbox.

7. Homework Helper

This is a very useful NLP project for all the parents out there who struggle with helping their children in completing complicated tasks assigned as homework to their kids. By designing a simple NLP-based app, you can help your kids with their homework.

8. Speech Recognition

Automatic Speech Recognition (ASR), or Speech to Text, is an NLP task that converts audio inputs into text. It is helpful for many applications, including automatic caption generation for videos, dictation to generate reports and other documents, or creating transcriptions of audio recordings.

9. Predictive Text (Sentence Autocomplete)

This project was conducted with the objective to automatically predict the text for the given input of text. It explores the way to predict sentences by using machine learning or deep learning models on the initial text input from users and predict the text by taking into consideration user-specific writing style.

10. Smart Assistant

Smart assistants like Apple's Siri and Amazon's Alexa recognize patterns in speech thanks to voice recognition, then infer meaning and provide a useful response. We now expect assistants like Alexa and Siri to understand contextual clues as they improve our lives and make certain activities easier like ordering items, and even

appreciate when they respond humorously or answer questions about themselves.

11. Social Media Monitoring

This project will help you to monitor your brand or company name and to find out where your audience is talking about you and distribute more content and ads there. Learn more about the type of audience that you have. See what your audience is saying about you and manage your reputation.

12. Online Search

Search engines use NLP to surface relevant results based on similar search behaviors or user intent, so the average person finds what they need without being a search-term wizard. For example, Google not only predicts what popular searches may apply to your query as you start typing, but it looks at the whole picture and recognizes what you're trying to say rather than the exact search words.

13. Analyzing Speech Emotions

In this project, the goal is to build a system that analyzes emotions in speech using any suitable dataset. It will help researchers and developers to better understand human emotions and develop applications that can recognize emotions in speech.

14. Language Identification

How often have you traveled to a city where you were excited to know what languages they speak? That's such a common thing. To discover a language, you don't always have to travel to that city, you might even come across a document while browsing through websites on the Internet or going through books in your library and may have the curiosity to know which language it is. This NLP project is all about quenching your curiosity only to build your own language identifier.

15. Essay Scoring

Automated Essay Scoring (AES) is an innovative application of NLP that has revolutionized educational assessment. AES systems utilize NLP to evaluate, and grade written essays based on various parameters like grammar, vocabulary, coherence, and argument structure.

16. Short Answer Grading

Automatic grading is not a new approach but the need to adapt the latest technology to automatic grading has become very important. As the technology has rapidly become more powerful on scoring exams, especially from the 1990s onwards, partially or wholly automated grading systems using computational methods have evolved and have become a major area of research. In particular, the demand of scoring of natural language responses has created a need for tools that can be applied to automatically grade these responses.

The student should first apply NLP techniques to the corpus of student answers for the purpose of measuring similarity between the student answers and the model answer. This is based on the number of common words. Then, the student should evaluate the relation between these similarities and marks awarded by scorers.

17. Question Generation

The goal of question generation is to generate a valid and fluent question according to a given passage and the target answer. Question generation can be used in many scenarios, such as automatic tutoring systems, improving the performance of question answering models and enabling chatbots to lead a conversation.

18. Detecting Paraphrases

This project is perfect for researchers and teachers who come across paraphrased answers in assignments. The student will work on building a system that identifies whether two sentences are paraphrases of each other or not. This project will also be helpful for researchers and developers as it will allow them to build systems that can recognize paraphrases.

19. Paraphrasing

Paraphrase generation is the task of generating an output sentence that preserves the meaning of the input sentence with variations in word choice and grammar. Two sentences are paraphrases if their meanings are equivalent, but their words and syntax are different.

Paraphrasing text can facilitate reading comprehension by transforming the text into a more familiar, and in the field of composition, allow writers to restate ideas from other works or their own drafts so that the reformatted language may better suit a voice, flow, or line of argument.

20. Title Generation

The aim of the project is to develop a generalized architecture that generates heading for the given text, which can be an article, a book's paragraph, newspaper contents, or any legal document. For this purpose, a model is proposed which uses an encoder-decoder recurrent neural network architecture consisting of LSTM and attention units.

21. Document Analysis (Topic Identification)

This is a very basic NLP Project that expects you to use NLP algorithms to understand them in depth. The task is to have a document and use relevant algorithms to label the document with an appropriate topic. A good application of this NLP project in the real world is using this NLP project to label customer reviews. The

companies can then use the topics of the customer reviews to understand where the improvements should be done on priority.



22. Spell Checkers (Autocorrect)

NLP-based spell checkers leverage advanced language models to understand the context in which words are used. This contextual understanding allows for more accurate error identification and correction suggestions, especially in cases of homophones or context- dependent spelling errors.

23. Market Intelligence

Leveraging Natural Language Processing (NLP) can be a game-changer for marketers seeking deeper insights into their customers, ultimately enhancing their ability to craft more impactful strategies. By analyzing elements such as topics, sentiment, keywords, and intent, marketers can significantly augment their market research efforts. This approach not only illuminates current trends but also unveils potential business opportunities. Moreover, the ability to discern customer pain points and monitor competitor activities through data analysis adds another layer of strategic advantage.

The incorporation of example of NLP in daily life in market intelligence not only refines the understanding of customer preferences but also empowers marketers to stay agile in

responding to dynamic market conditions, thus contributing to more informed and effective decision-making processes.

24. Resume Parsing System (Hiring and Recruitment)

The Human Resource department holds a crucial role in every organization, tasked with the pivotal responsibility of selecting the most suitable employees. In today's highly competitive environment, recruiters often face the daunting task of reviewing numerous resumes, sometimes numbering in the hundreds or thousands, for a single position. This process of filtering resumes and shortlisting candidates can be time-consuming. However, there is a solution – automation through natural language processing (NLP).

This automated approach ensures unbiased resume filtering, facilitating the selection of the most qualified candidates for a vacant position, all without extensive manual effort. Many companies employ Application Tracking Systems (ATS) to efficiently screen resumes and enhance the overall recruitment process.



25. Recommender System

The recommendation engine is based on the input given by the pre-trained Natural Language Processing model, where the items from the dataset are segregated into broad themes based on the product description. These themes are then put into clusters based on their similarity.

26. Disease Diagnosis (Healthcare)

Natural Language Processing (NLP) can be used for diagnosing diseases by analyzing the symptoms and medical history of patients expressed in natural language text. NLP techniques can help in identifying the most relevant symptoms and their severity, as well as potential risk factors and comorbidities that might be indicative of certain diseases.

27. Conversion from Unstructured Data into Structured Data

NLP techniques can be used to convert the unstructured data into structured data. This student should extract the unstructured data from business cards, medical reports, ... etc. and convert them into useful structured form. You can use NLP to identify specific patterns or signals within the text data that might be indicative of a particular condition.

28. Custom Project

The student can choose his/her own project. This can be done after a formal approval from the instructor.